As a low temperature of mixing indicates a stable, valuable structure for the alloy, the objective was to find compositions mixing temperatures as low as possible. But there is one bottleneck: determining this mixing temperature for a given alloy generally takes days. The question that we set out to solve algorithmically is similar: given the data set that we see, what is the next composition we should experiment with (in terms of how much lead, tin, germanium, and manganese should be present) to find the minimum temperature of mixing?

Another example is in mining and oil drilling, or more specifically finding the region within a big area that has the highest yield of valuable minerals or oil. This involves extensive planning, investment, and labor—again an expensive undertaking. As digging operations have significant negative impacts on the environment, there are regulations in place to reduce mining activities, placing a limit on the number of "function evaluations" that may be done in this optimization problem.

The central question in expensive black-box optimization is: what is a good way to decide where to evaluate this objective function, so that its optimum may be found at the end of the search? As we will see in a later example, simple heuristics such as random or grid search, which are approaches implemented by popular Python packages such as Scikit-learn, may lead to wasteful evaluations of the objective function and thus overall poor optimization performance. This is where Bayesian optimization comes into play.

1.2 Introducing Bayesian optimization

With the problem of expensive black-box optimization in mind, we will now introduce Bayesian optimization as a solution to this problem in this section. This will give you a high-level idea of what Bayesian optimization is and how it leverages probabilistic machine learning to optimize expensive, black-box functions.

IMPORTANT **Definition of Bayesian optimization**

Bayesian optimization is a machine learning technique that simultaneously maintains a predictive model to learn about the objective function and makes decisions about how to acquire new data to refine our knowledge about the objective.

In a Bayesian optimization procedure, we make decisions based on the recommendation of a Bayesian optimization algorithm. Once we have taken the Bayesian optimization-recommended action, the Bayesian optimization model gets updated based on the result of that action, and proceeds to recommend the next action to take. This process repeats until we are confident that we have zeroed in on the optimal action.

There are two main components to this workflow:

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